

Example: Bakery

Problem Statement

A Bakery produces both, breads and cakes in batches. A batch of bread contributes \$50 to profit and a batch of cake \$60. The bakery can sell at most 90 batches of bread each week and no more than 60 batches of cake.

Production is mainly constrained by the available oven time (a batch of bread requires 1 hour of baking, a batch of cake requires 45 minutes) and the manual labor associated with a batch (30 minutes for bread, 60 minutes for cake). Each week, 4200 minutes of labor are available and 4650 minutes of oven time.

What are the optimal production quantities?



Bread (B)

Profit: \$50 / batch

- **Max Demand:** 90 batches/wk
- **Oven Time:** 1 hour (60 min)
- **Labor:** 30 minutes



Cake (C)

Profit: \$60 / batch

- **Max Demand:** 60 batches/wk
- **Oven Time:** 45 minutes
- **Labor:** 60 minutes

Constraint Analysis

Labor Limit

$$30B + 60C \leq 4,200 \text{ mins}$$

Y-Intercept (Cake Only)

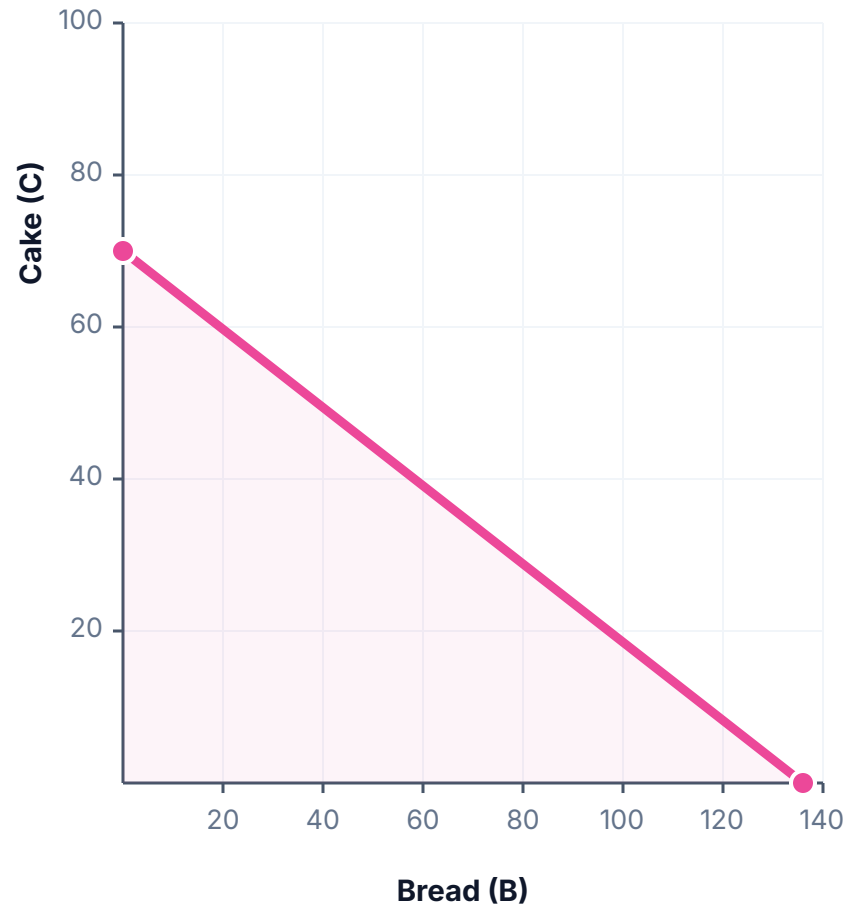
Max 70 Cakes

4,200 mins / 60 mins per cake

X-Intercept (Bread Only)

Max 140 Breads

4,200 mins / 30 mins per bread

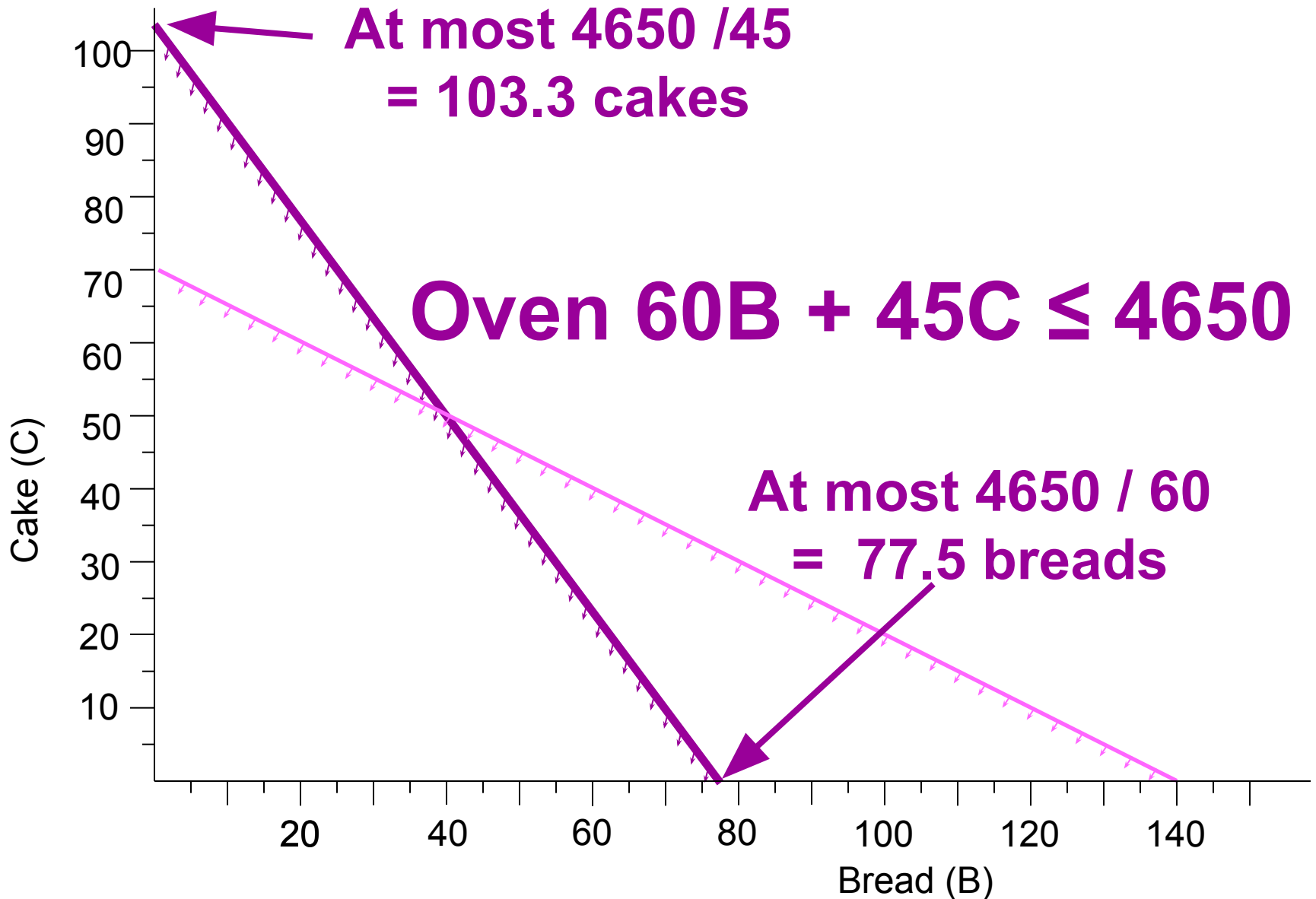


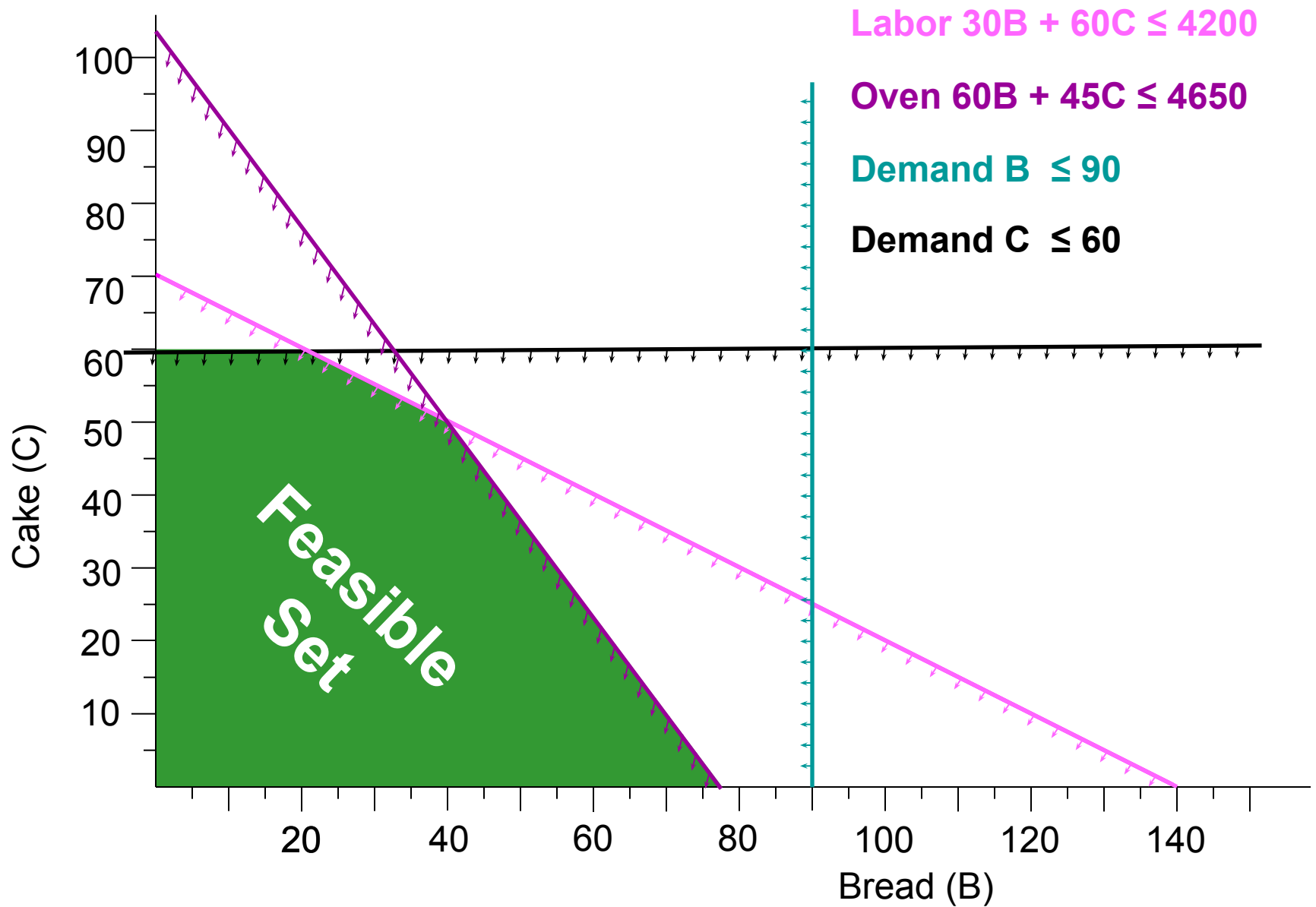
$$\text{Labor } 30B + 60C \leq 4200$$

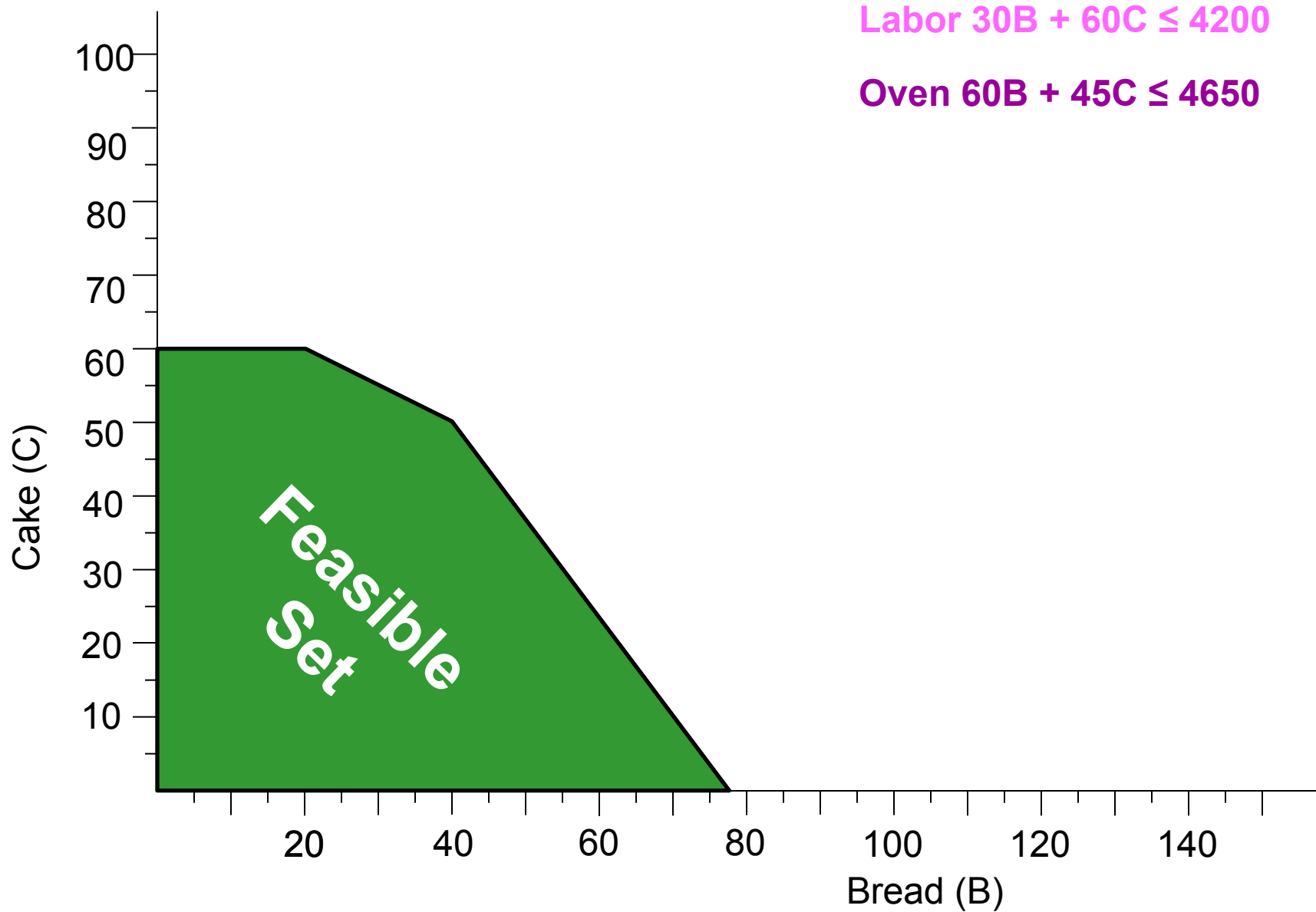
**At most $4650 / 45$
 $= 103.3$ cakes**

$$\text{Oven } 60B + 45C \leq 4650$$

**At most $4650 / 60$
 $= 77.5$ breads**







**Can we make
\$ 1,500?**

Labor $30B + 60C \leq 4200$

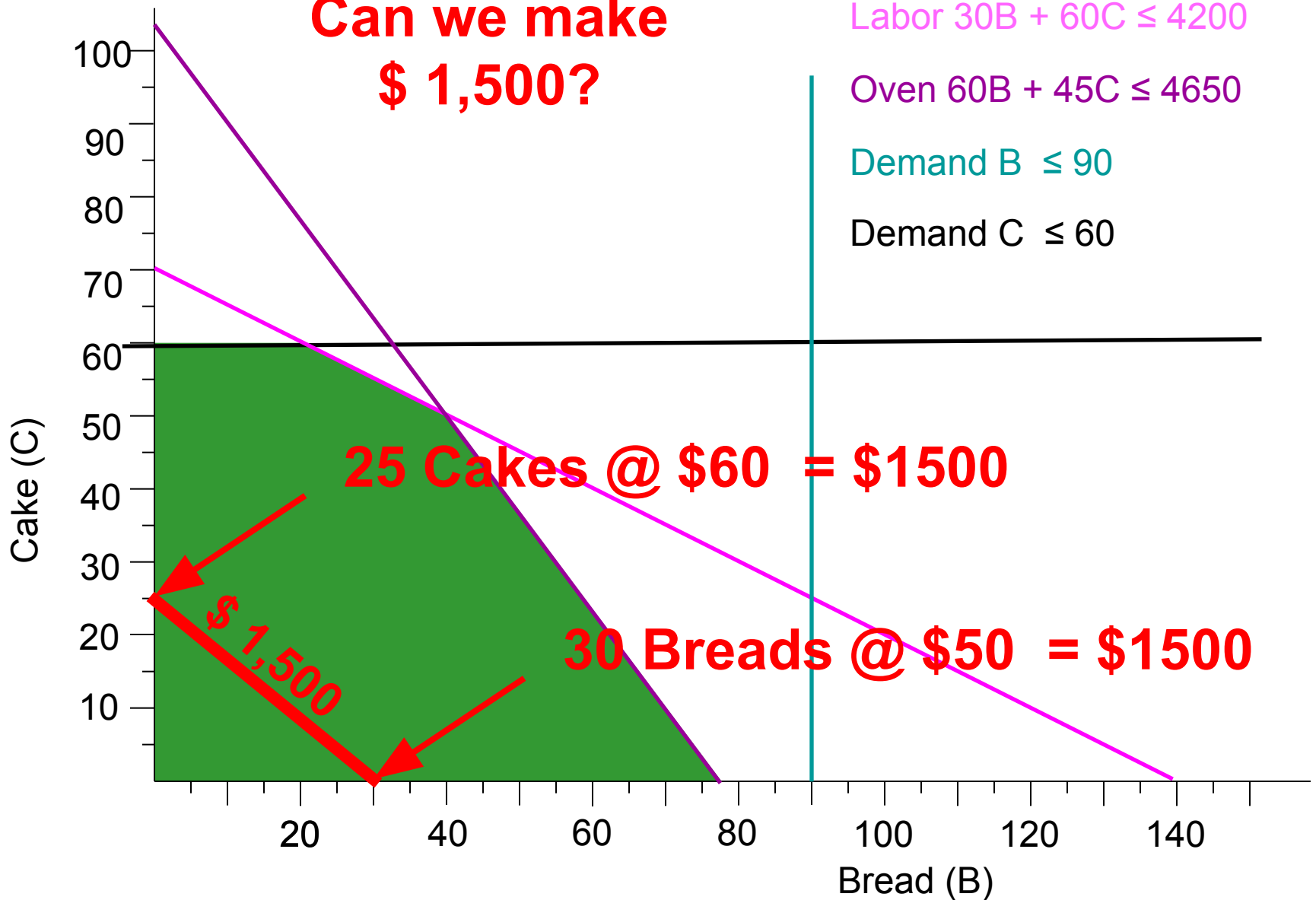
Oven $60B + 45C \leq 4650$

Demand B ≤ 90

Demand C ≤ 60

25 Cakes @ \$60 = \$1500

30 Breads @ \$50 = \$1500



**Can we make
\$ 3,000?**

$$\text{Labor } 30B + 60C \leq 4200$$

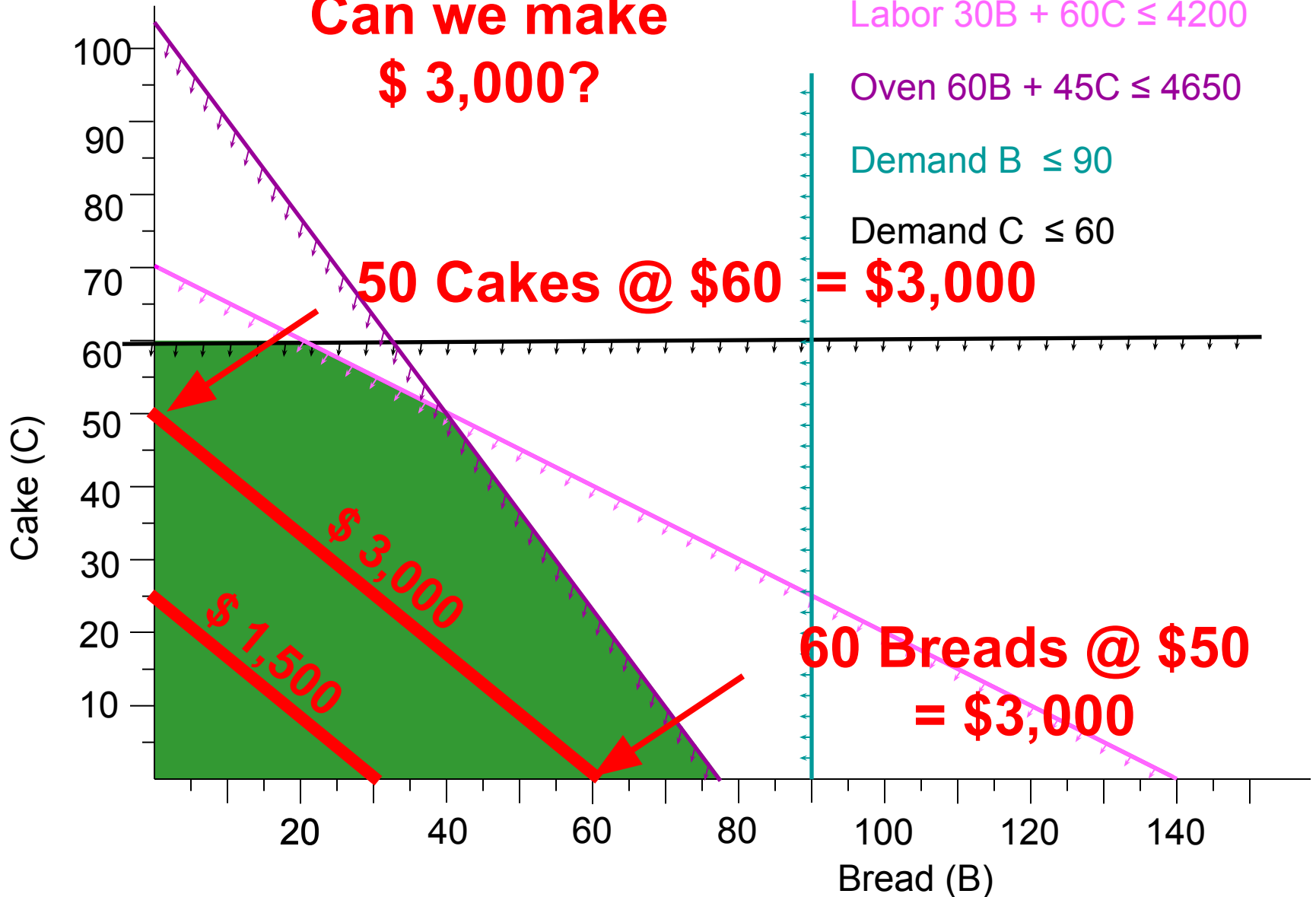
$$\text{Oven } 60B + 45C \leq 4650$$

$$\text{Demand } B \leq 90$$

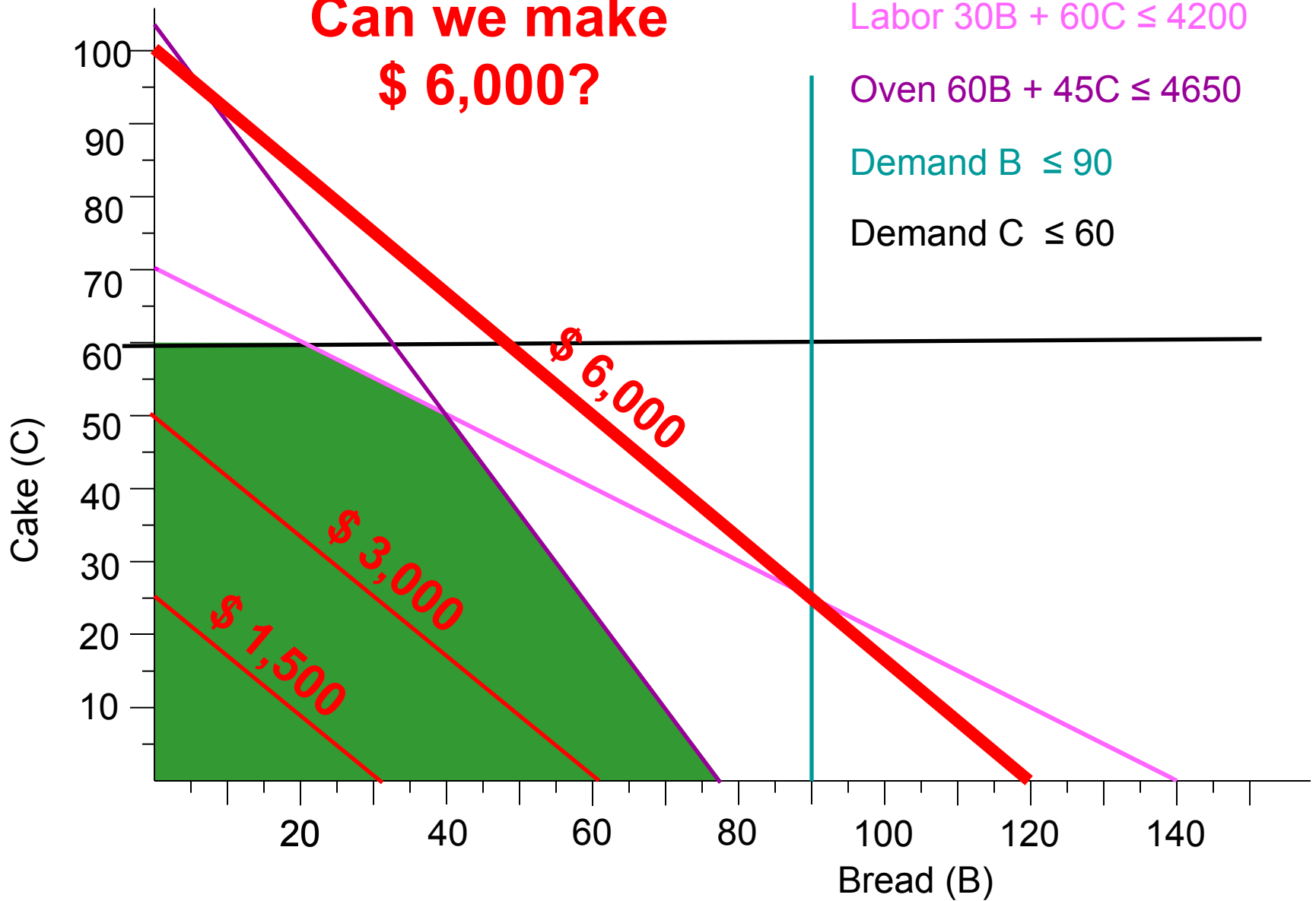
$$\text{Demand } C \leq 60$$

50 Cakes @ \$60 = \$3,000

60 Breads @ \$50 = \$3,000



**Can we make
\$ 6,000?**



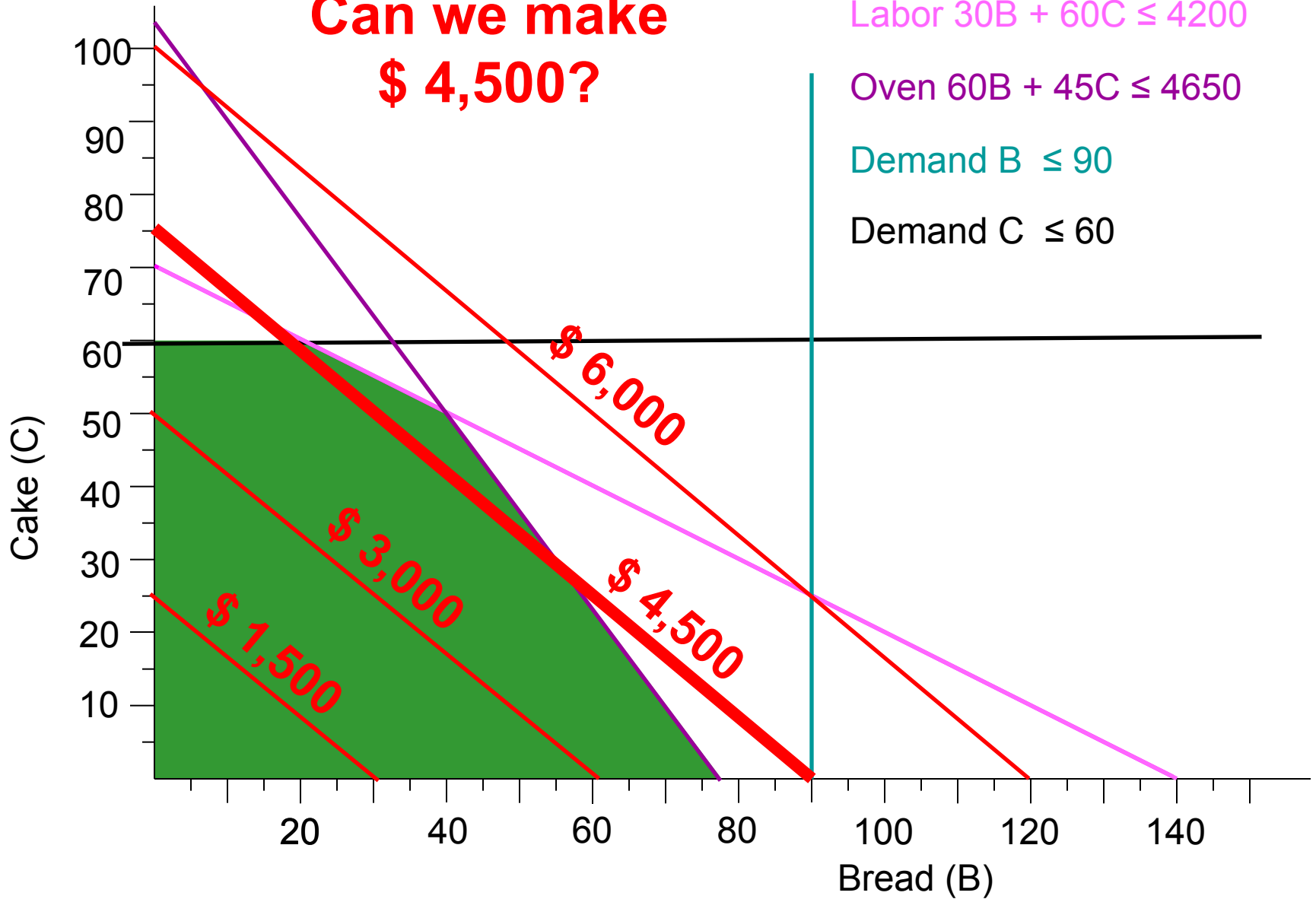
**Can we make
\$ 4,500?**

Labor $30B + 60C \leq 4200$

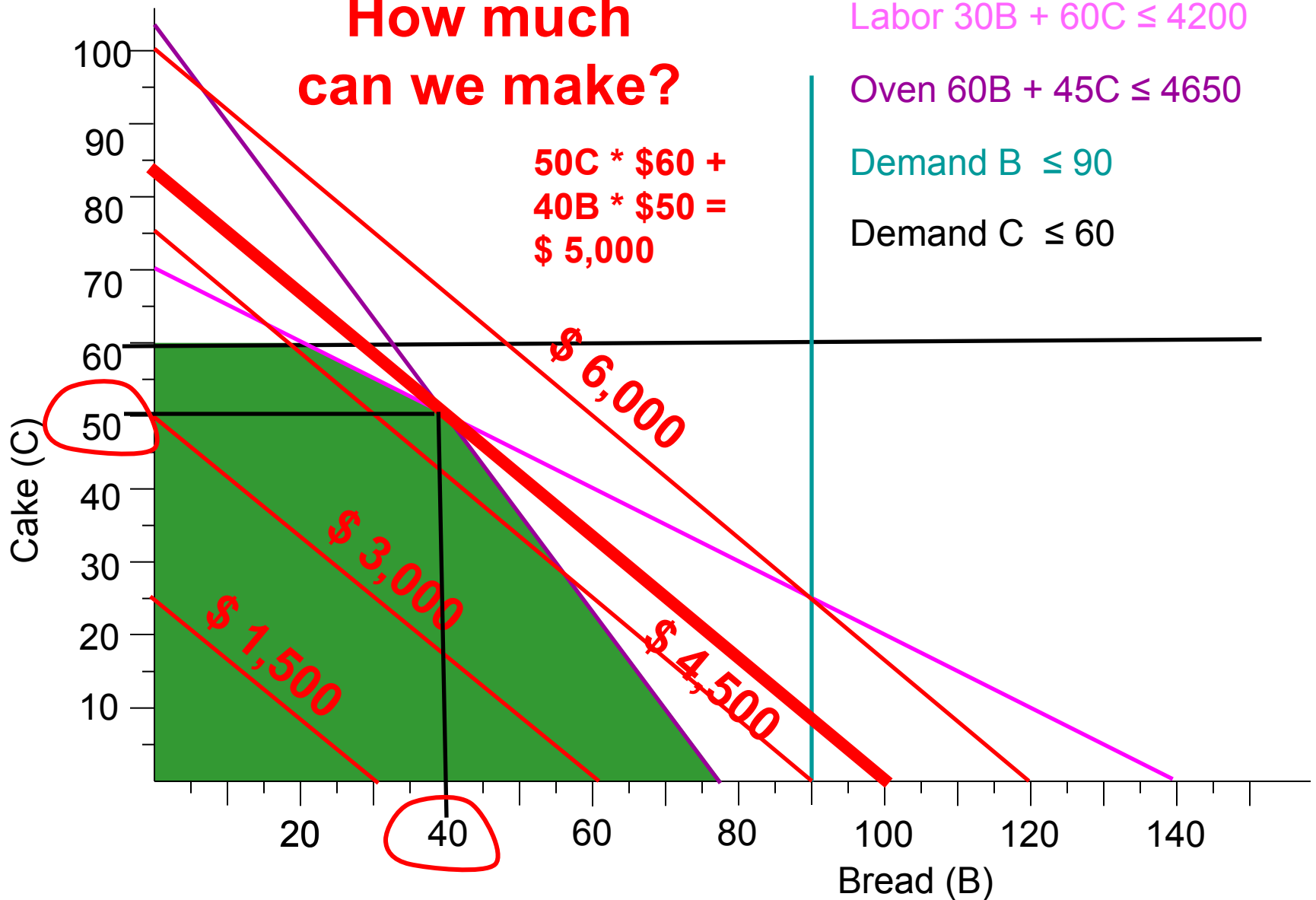
Oven $60B + 45C \leq 4650$

Demand B ≤ 90

Demand C ≤ 60



How much can we make?



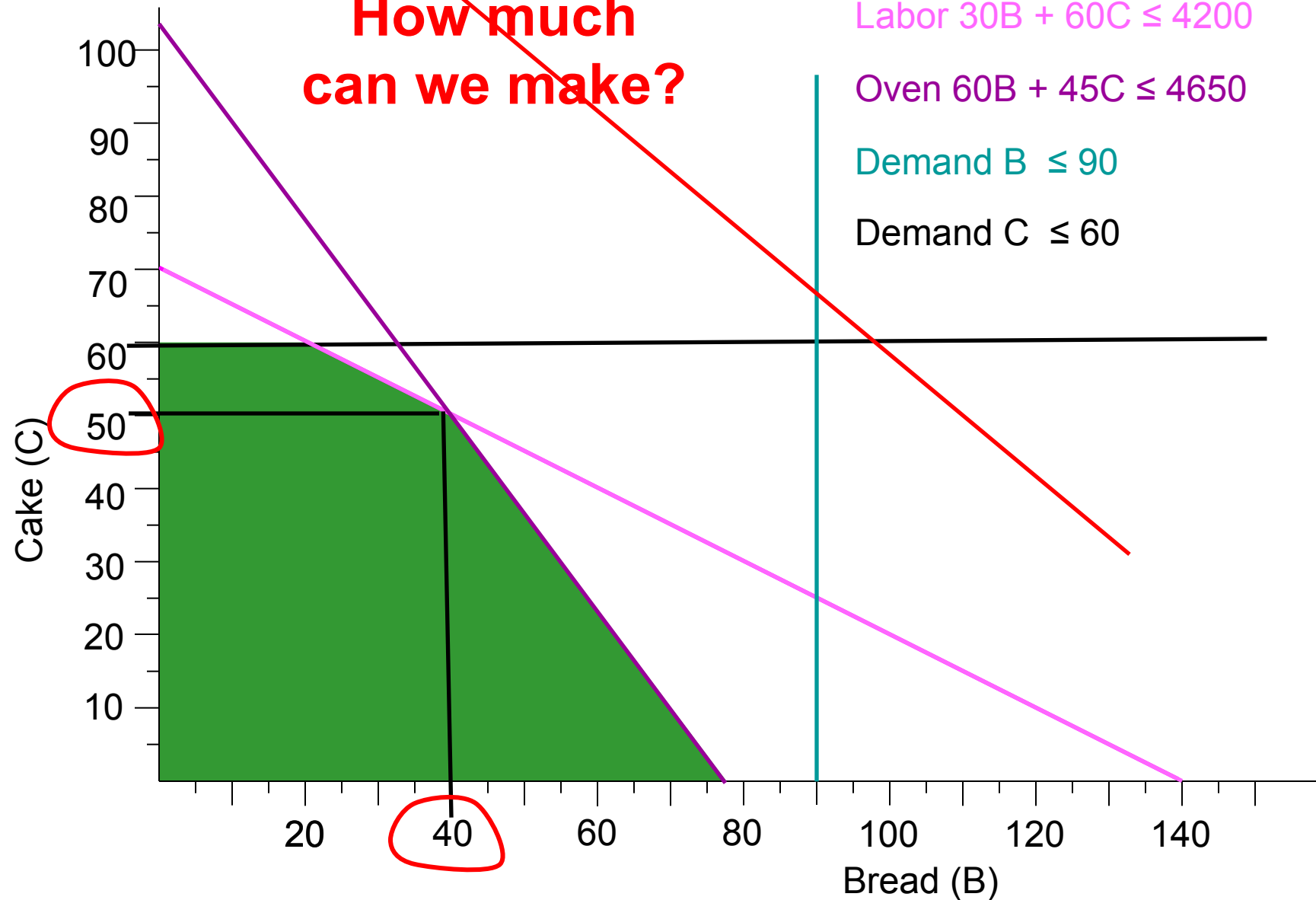
**How much
can we make?**

Labor $30B + 60C \leq 4200$

Oven $60B + 45C \leq 4650$

Demand B ≤ 90

Demand C ≤ 60



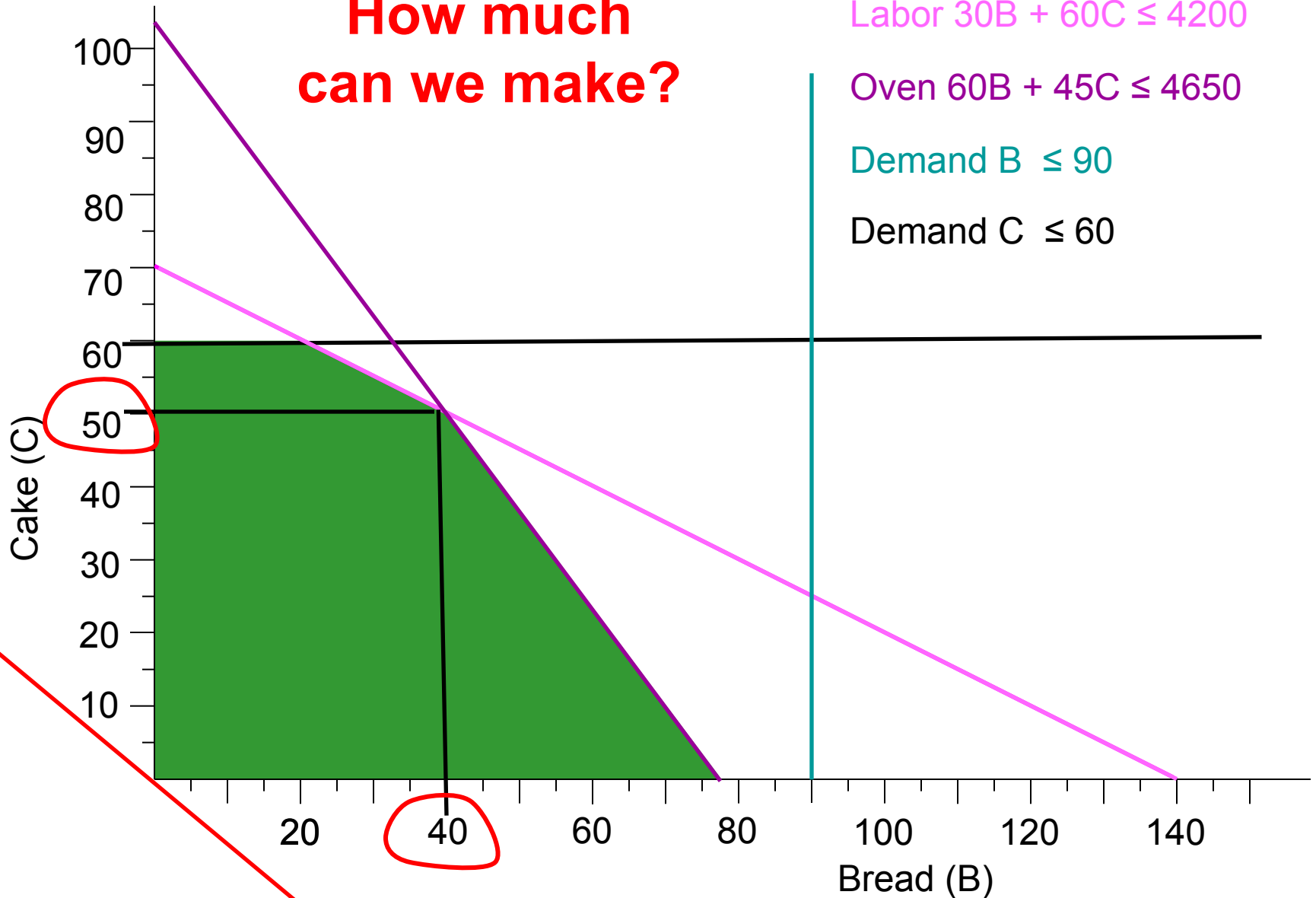
**How much
can we make?**

Labor $30B + 60C \leq 4200$

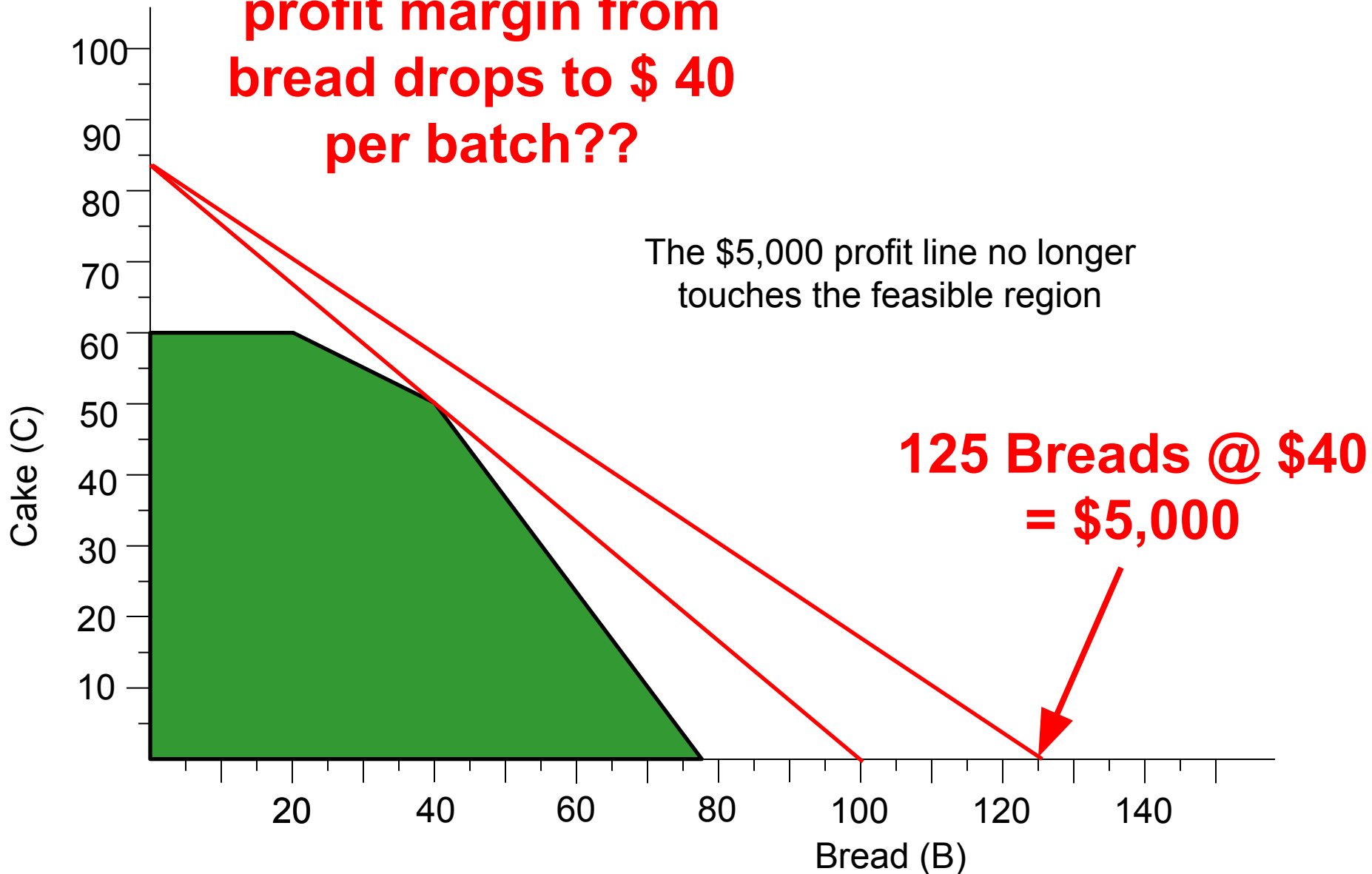
Oven $60B + 45C \leq 4650$

Demand B ≤ 90

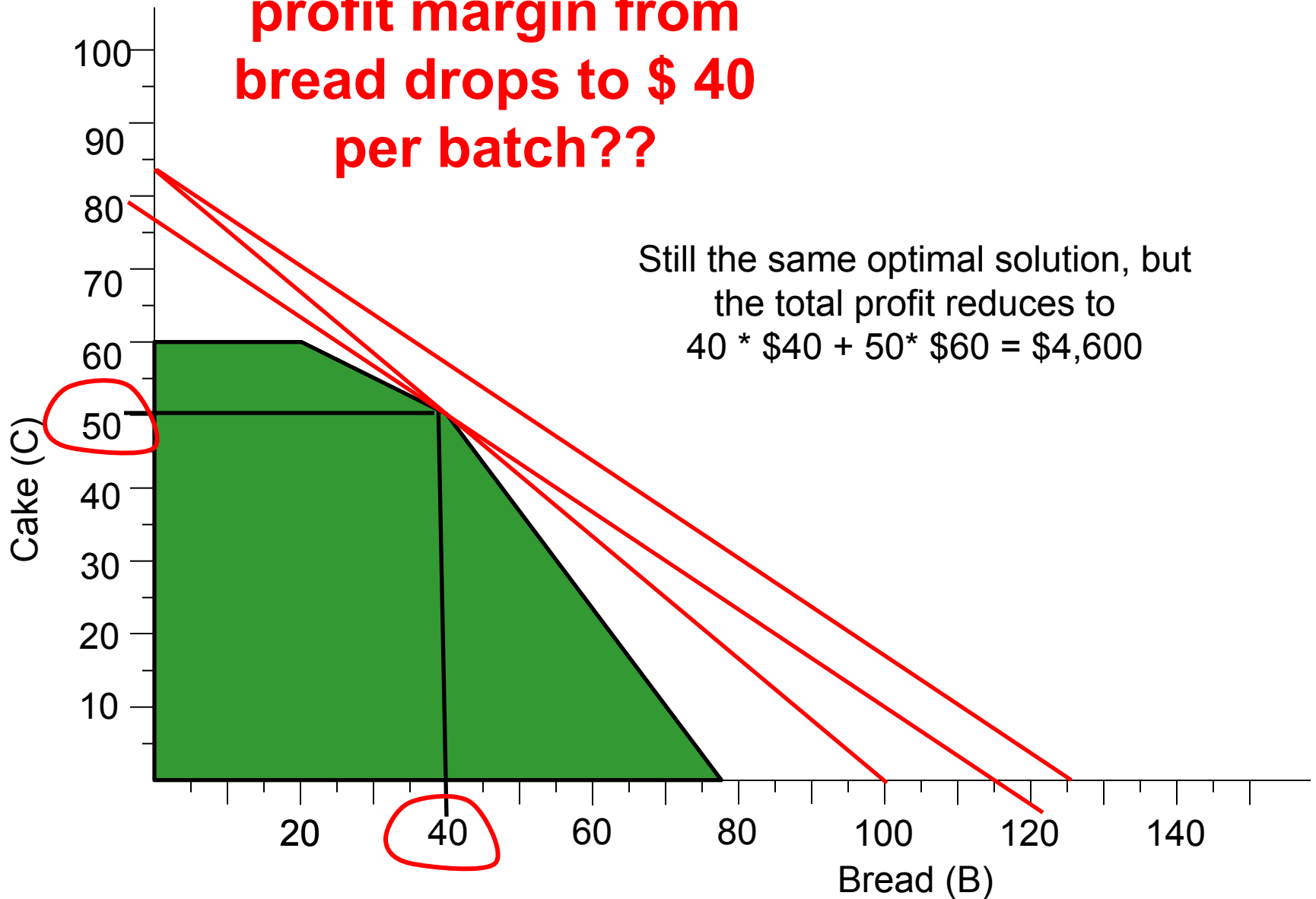
Demand C ≤ 60



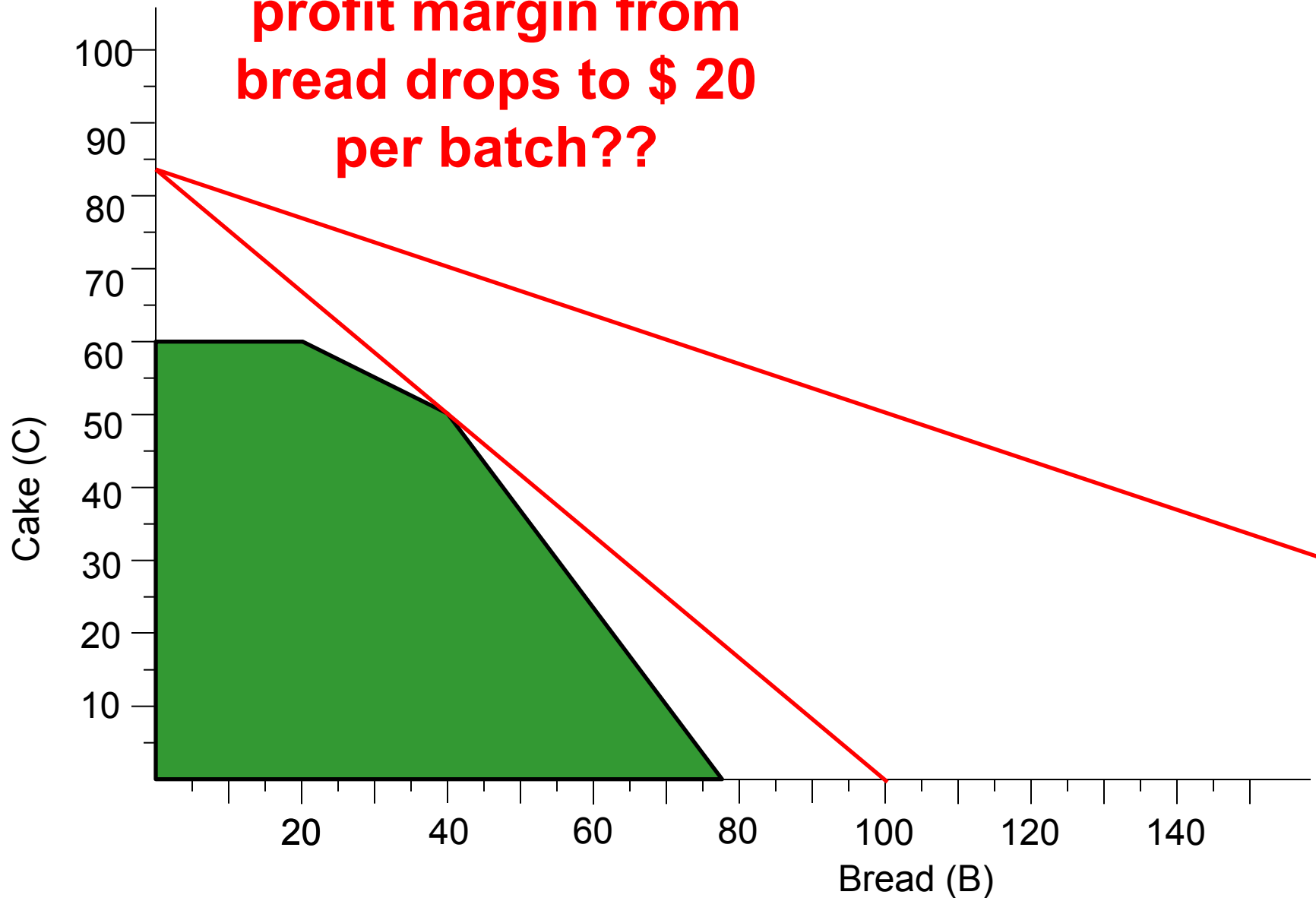
**What happens if the
profit margin from
bread drops to \$ 40
per batch??**



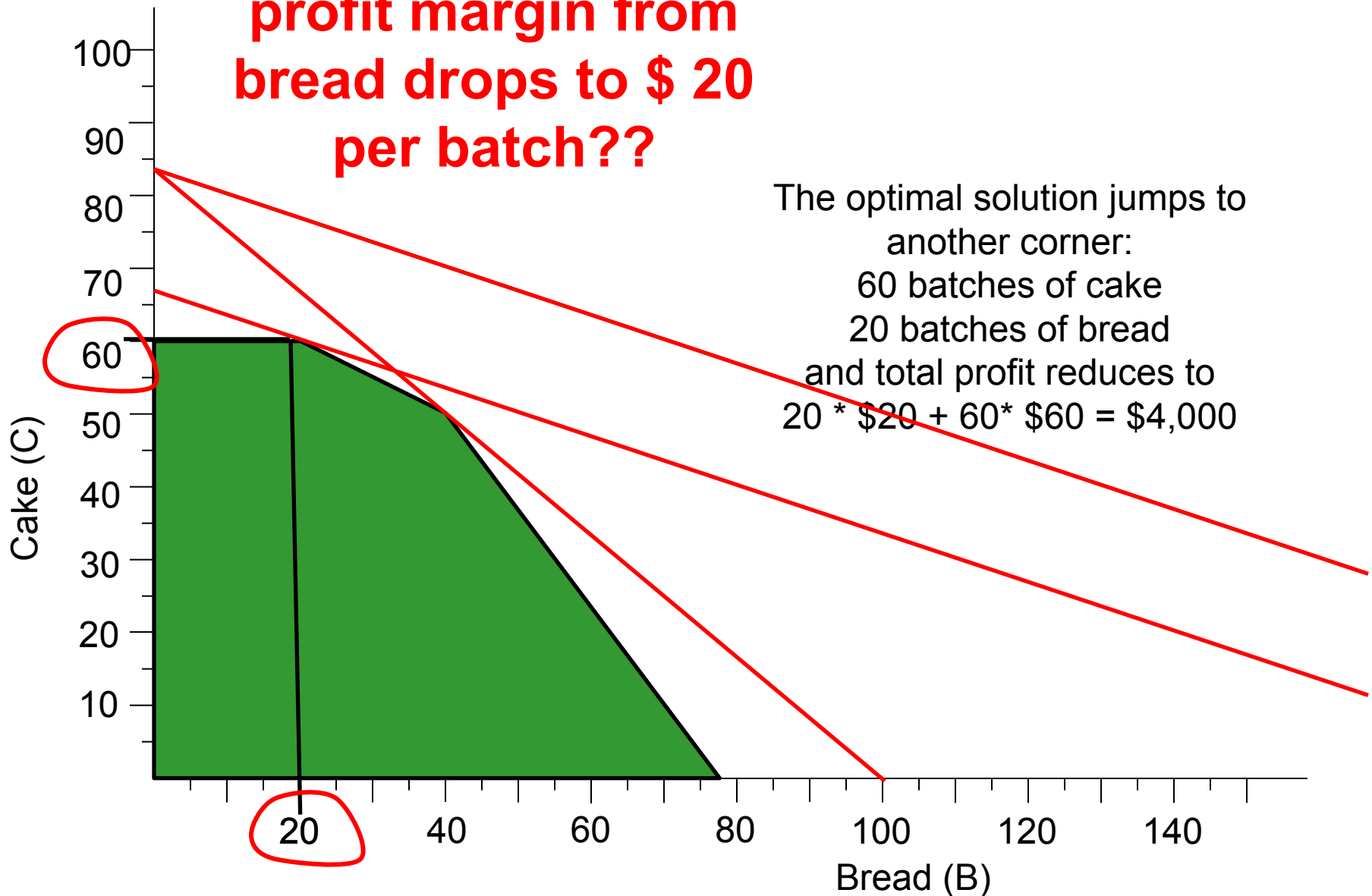
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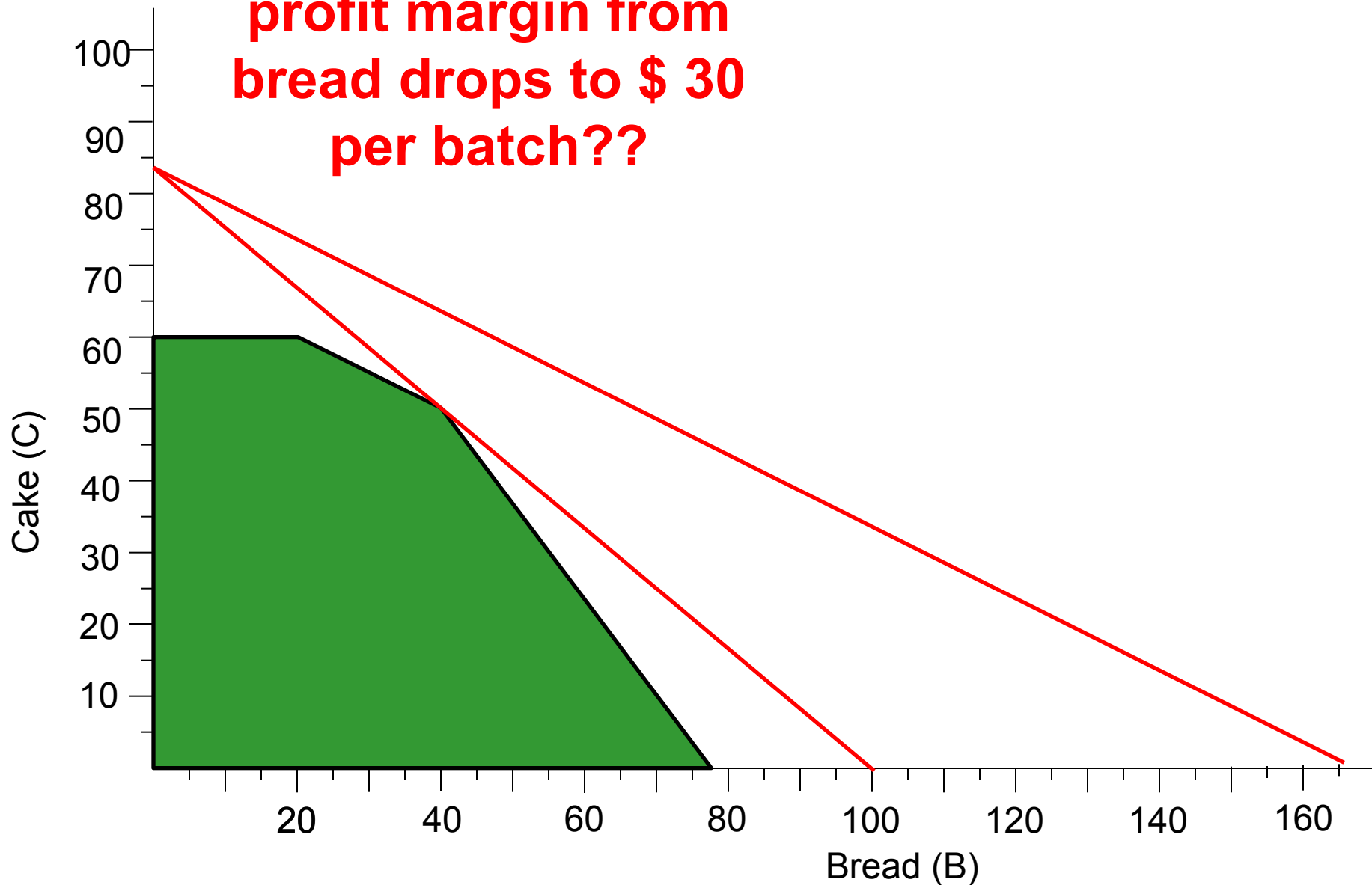
**What happens if the
profit margin from
bread drops to \$ 20
per batch??**



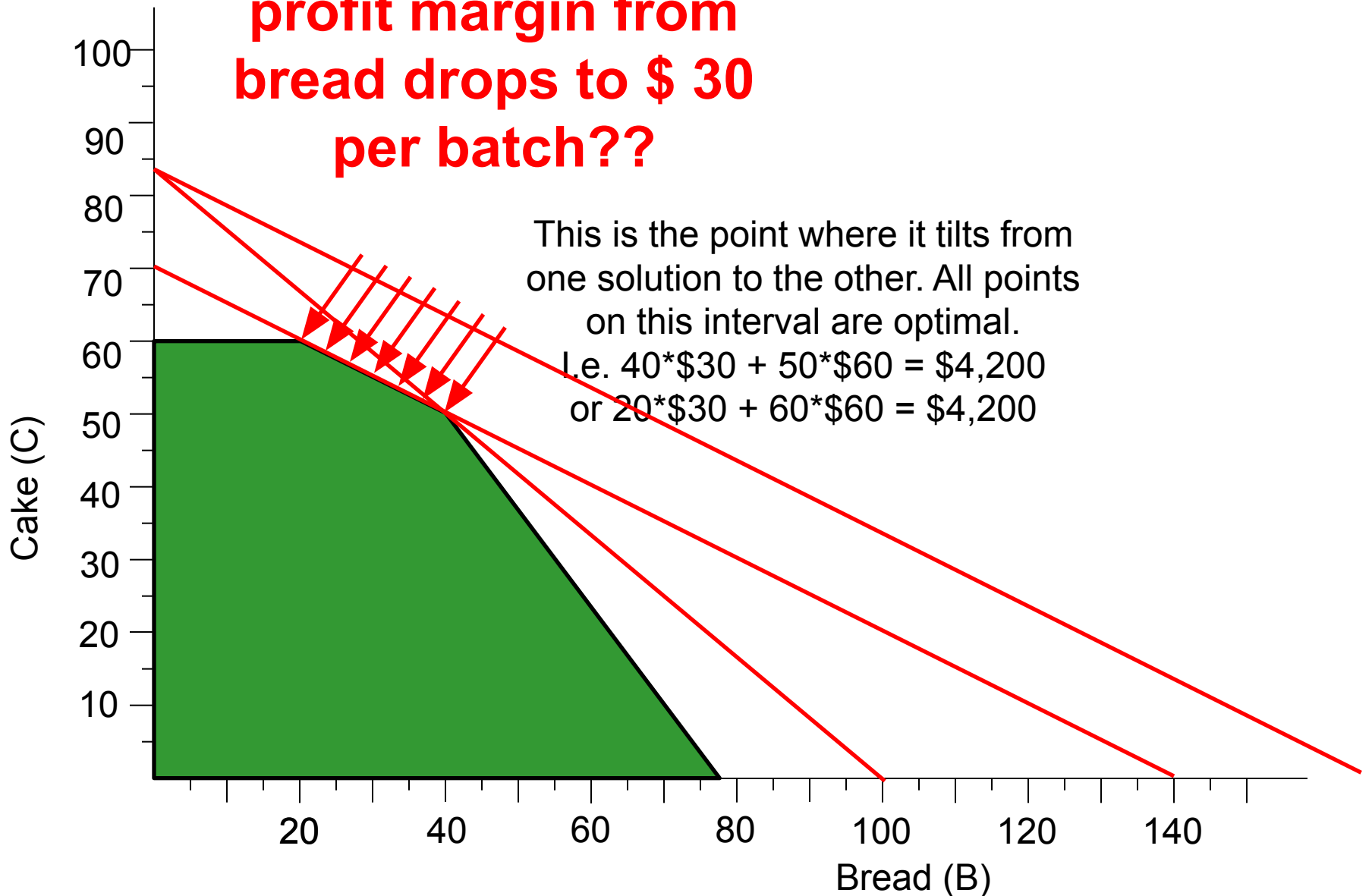
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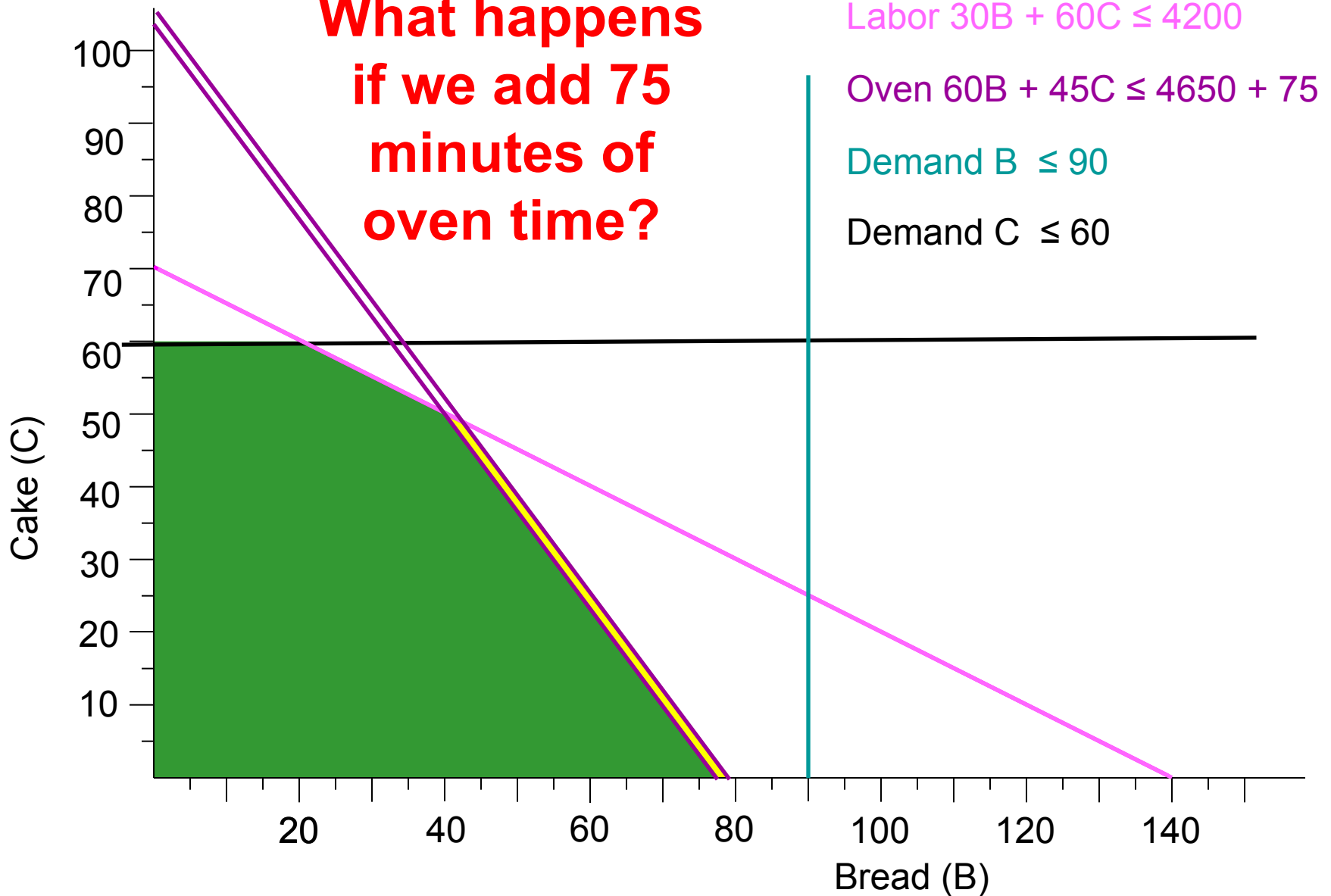
**What happens if the
profit margin from
bread drops to \$ 30
per batch??**



**What happens if the
profit margin from
bread drops to \$ 30
per batch??**



**What happens
if we add 75
minutes of
oven time?**



**What happens
if we add 75
minutes of
oven time?**

$$\text{Labor } 30B + 60C \leq 4200$$

$$\text{Oven } 60B + 45C \leq 4650 + 75$$

$$\text{Demand } B \leq 90$$

$$\text{Demand } C \leq 60$$

The optimal solution changes to :
 $42 * \$50 + 49 * \$60 = \$5,040$

